

Reproducibility and Progress in Recommender Systems

Doctoral Thesis

Faisal Shehzad

Department of Artificial Intelligence and Cybersecurity
University of Klagenfurt, Austria

Reviewer:	Title. Full Name
Second reviewer:	Full Name
Supervisor:	Univ.-Prof. Dr. Dietmar Jannach
Co-Supervisor:	Assoc.-Prof. Dr. Konstantin Schekotihin

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Acknowledgments

“Gratitude makes sense of our past, brings peace for today, and creates a vision for tomorrow.” — Melody Beattie

I would like to express my sincere gratitude to all those who supported me throughout my doctoral journey. Completing this thesis has been a long and challenging process, but it has also been a very valuable and rewarding experience. I am truly thankful to everyone who helped me stay motivated and continue working toward this important goal.

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“Alone we can do so little; together we can do so much.” — Helen Keller

List of Publications included in this thesis

1. Faisal Shehzad and Dietmar Jannach. “Everyone’s a Winner! On Hyperparameter Tuning of Recommendation Models.” Proceedings of the 17th ACM Conference on Recommender Systems. 2023.
2. Faisal Shehzad and Dietmar Jannach. “Performance Comparison of Session-based Recommendation Algorithms based on GNNs.” European Conference on Information Retrieval. Cham: Springer Nature Switzerland, 2024.
3. Faisal Shehzad, Maurizio Ferrari Dacrema and Dietmar Jannach. “A Worrying Reproducibility Study of Intent-Aware Recommendation Models.” Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval. 2025.
4. Faisal Shehzad and Dietmar Jannach. “Revisiting the Performance of Graph Neural Networks for Session-based Recommendation.” Proceedings of the Nineteenth ACM Conference on Recommender Systems. 2025.
5. Faisal Shehzad, Timo Breuer, Maria Maistro and Dietmar Jannach. “We Share Our Code Online”: Why This is Not Enough to Ensure Reproducibility and Progress in Recommender Systems Research.” Proceedings of the Nineteenth ACM Conference on Recommender Systems. 2025.
6. Faisal Shehzad and Dietmar Jannach. “Reproducibility of LLM-based Recommender Systems Research: Worrying Observations from an Initial Analysis.” Proceedings of the 34th ACM Conference on User Modeling, Adaptation and Personalization. 2026.

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List of Abbreviations

η^2	Effect Size
R	Recall
AMZm	Amazon Digital Music
API	Application Programming Interface
BIGCF	Bilateral Intent-guided Graph Collaborative Filtering
CIKM	The ACM International Conference on Information and Knowledge Management
ConvMF	Convolutional Matrix Factorization
Cov	Coverage
DCCF	Disentangled Contrastive Collaborative Filtering
DGCF	Disentangled Graph Collaborative Filtering
DIGI	Diginetica
DL	Deep Learning
EASE ^R	Embarrassingly Shallow Autoencoder
GMF	Generalized Matrix Factorization
GNN	Graph Neural Network
GRU4Rec	Gated Recurrent Unit for Recommendations
IARS	Intent Aware Recommender System
ICDM	The IEEE International Conference on Data Mining
ICLR	The International Conference on Learning Representations
IDS4NR	Intent Disentanglement and Feature Self-supervision
KDD	The ACM SIGKDD Conference on Knowledge Discovery and Data Mining
KGIN	Knowledge Graph-based Intent Network
LLM	Large Language Model
MAP	Mean Average Precision
MF	Matrix Factorization
ML-1M	MovieLens-1M
MRR	Mean Reciprocal Rank
Mult-DAE	Multinomial Likelihood Denoising Autoencoder

Mult-VAE	Multinomial Likelihood Variational Autoencoder
NARM	Neural Attentive Recommendation Machine
NDCG	Normalized Discounted Cumulative Gain
NeuMF	Neural Network-based Collaborative Filtering
NGCF	Neural Graph Collaborative Filtering
ONCF	Outer Product-based Neural Collaborative Filtering
P-Time	Prediction Time
Pop	Popularity
RecSys	The ACM Conference on Recommender Systems
RETAIL	Retailrocket
RS	Recommender System
RSC15	RecSys Challenge 2015
SBRS	Session-based Recommender System
SIGIR	The ACM SIGIR Conference on Research and Development in Information Retrieval
SR	Sequential Rules
SR-GNN	Session-based Recommendation with Graph Neural Networks
STAMP	Short-Term Attention/Memory Priority Model
T-Time	Training Time
URL	Uniform Resource Locator
UserKNN	User-based Nearest-Neighbor Method
WSDM	The ACM International Conference on Web Search and Data Mining
WWW	The ACM Web Conference

Abstract

Recommender systems have become an integral part of modern digital platforms, helping users navigate large collections of products, services, and content. Over the past decade, research in this field has increasingly focused on sophisticated models designed by using machine learning and deep learning approaches. These advances are frequently accompanied by claims of improved performance over existing methods. However, whether such improvements represent genuine progress depends critically on the reproducibility of the experiments and the fairness of the underlying comparisons.

This thesis investigates the reproducibility and reliability of empirical progress in recommender-systems research. Across [six](#) studies, we examine how methodological choices, baseline selection, hyperparameter optimization, and artifact-sharing practices influence conclusions about the effectiveness of newly proposed models. The analyses cover traditional top-n recommendation, session-based recommendation, and intent-aware recommender systems.

Our findings reveal several recurring methodological concerns. First, hyperparameter optimization is often insufficiently documented, particularly for baseline models, and experiments demonstrate that selective tuning can substantially alter model rankings and make almost any model appear superior to inadequately optimized competitors. Second, controlled evaluations of session-based recommendation models show that carefully tuned k-nearest neighbors and well established simpler neural approaches such as GRU4Rec can match or outperform substantially more complex graph neural networks. We further demonstrate that experimental choices such as embedding size, random seeds, and optimization on test data can markedly affect reported results. Similar observations emerge for intent-aware recommendation: when recent models are reproduced using the artifacts provided by their authors and compared with systematically tuned traditional baselines, at least one traditional method consistently achieves comparable or better performance.

In the last study, an analysis of 65 recent recommender-systems papers published at a high ranked venue shows that, although sharing source code for newly proposed models has become common, the artifacts required to reproduce complete experiments—including baseline implementations, preprocessing procedures, data splits, and hyperparameter-tuning details—are frequently missing. Thus, the provided artifacts may allow us to reproduce the claims made about the proposed models, but the central claim that these models outperform the considered baselines remains unverifiable.

Overall, this thesis demonstrates that increasing model complexity does not necessarily translate into reliable empirical progress. It highlights the importance of strong and diverse baselines, systematic optimization, transparent experimental procedures, and sharing of complete reproducibility package for establishing credible advances in recommender-systems research.

1. Introduction and Background

Digital platforms provide users with an increasingly large volume of products, services, and content, making it difficult to identify the options that are most relevant to their needs. Recommender systems (RS) address this information-overload problem by analyzing user preferences, previous interactions, contextual information, and item characteristics to generate personalized suggestions that support decision-making and improve access to relevant content [1, 2]. They are now widely deployed across domains such as e-commerce, entertainment, social media, news, e-learning, tourism, finance, and healthcare [3, 4, 5, 6, 7, 8]. In addition to improving user experience, RS generate substantial organizational value by facilitating product discovery, increasing user engagement, supporting customer retention, and contributing to business growth [9, 10]. Their broad applicability and growing societal and economic influence have therefore established RS as a major research area.¹

To operationalize personalized recommendation, a wide range of RS algorithms has been proposed in the literature. A major line of research focuses on top-n recommendation, where algorithms generate a ranked list of items that are expected to be relevant to a user. Many widely studied top-n approaches rely on collaborative filtering [11, 12]. Early top-n RS primarily relied on heuristic methods and matrix-factorization techniques [11, 13]. More recent approaches have increasingly incorporated neural architectures, which have been reported to improve recommendation performance in various settings [14, 15, 16, 17, 18]. These models typically exploit users’ historical interactions to learn their preferences and generate personalized ranked lists of recommended items.

However, access to users’ long-term interaction histories is not always available in real-world applications, for example because of privacy considerations or because users interact with a service without logging in [19]. In such situations, the available interactions are typically limited to a short period of activity, commonly referred to as a session. Session-based recommender systems (SBRS) have therefore been developed to generate recommendations from the sequence of interactions observed within the current session, without access to the user’s historical profile [20, 21, 22].

Since an anonymous session contains only a limited sequence of interactions, most SBRS approaches model the order and recurrence of actions within that sequence. Early methods relied on identifying frequent patterns mining, Markov models and Markov decision processes [23, 24, 25]. Recently, advances in neural architectures have substantially transformed the field of SBRS [26]. As a result, many modern SBRS methods rely on neural sequence-learning techniques. A landmark contribution in this area is GRU4Rec [27], which applies recurrent neural networks to model sequential dependencies within sessions and remains a widely used baseline for evaluating newly proposed methods. Building on this work, researchers have introduced increasingly sophisticated neural architectures, including attention-based models, transformer-based approaches, and many more, to better capture complex and

¹I draft all the content myself and use ChatGPT only to improve the language, grammar, and readability.

evolving patterns in session data [28, 29, 30]. More recently, graph neural networks (GNNs) have received increasing attention in SBRS because of their success in modeling complex relational structures in applications such as traffic-flow prediction, dynamic routing and navigation, anomaly detection, and public transport network utilization [31, 32]. GNN-based SBRS approaches typically represent the interactions within a session as a graph and learn item representations by propagating information across connected items. The literature frequently reports that such models achieve strong performance and outperform GRU4Rec and other established neural baselines. Representative examples of these models include [33, 19, 34, 35, 36, 37, 38].

Top-n and session-based approaches differ in the interaction history available to the model; however, neither setting necessarily identifies the purpose underlying a user’s current behaviour [17, 18, 39, 40, 28, 19, 34, 35]. This limitation has motivated research on intent-aware recommender systems (IARS). Top-n recommender systems typically consider a user’s entire interaction history when generating a recommendation list [18, 17, 41]. Consequently, the resulting recommendations may reflect a broad range of interests accumulated over time rather than the user’s immediate needs. However, users often access online platforms with a specific goal or intent in mind. For example, when using a video-streaming platform, a user may wish to find a family-oriented film for a weekend evening, search for content relevant to a work-related task, or simply explore the available catalogue for inspiration [42, 21]. In contrast to top-n approaches, SBRS [19, 34, 35] generate recommendations primarily from a user’s recent interactions within an ongoing session. Nevertheless, relying solely on either long-term or short-term interaction histories may not be sufficient to produce optimal recommendations.

To address these limitations, IARS have emerged as an important research direction. These systems aim to infer and incorporate users’ explicit or latent intents into the recommendation process. Intent-aware approaches have been proposed for several recommendation scenarios, including traditional top-n [43, 44, 45, 46, 47] and sequential recommendation [48, 49, 50]. In this thesis, however, we included only those IARS designed for traditional top-n recommendation settings.

Taken together, developments in top-n, SBRS, and IARS reflect a broader shift from heuristic and latent-factor methods toward increasingly sophisticated neural and graph-based architectures designed to capture richer patterns in user behavior. While this evolution has substantially expanded the modeling capacity of RS, greater architectural complexity does not necessarily translate into reliable empirical progress. This raises a central question: **do the improvements reported for recent models persist when these models are independently reproduced and compared with strong, systematically tuned baselines under consistent experimental conditions?** Recent reproducibility studies have shown that simpler models developed more than a decade ago can match or outperform recent deep learning (DL) models for top-n recommender systems when properly tuned [11, 51, 52, 53, 54]. Several recent studies reassessed the effectiveness of a fifteen-year-old matrix-factorization-based model and found that it still outperformed several models currently considered state-of-the-art [55, 56, 57]. These issues are not only limited to

top-n recommendation. Numerous reproducibility studies in sequential recommendation [20, 58, 59] have similarly demonstrated that simpler models can achieve competitive performance and, in many cases, outperform more complex approaches.

Beyond baseline selection and hyperparameter optimization, several experimental choices can further contribute to unreliable assessments of progress. One such choice concerns the embedding dimension. Previous studies [60] have observed that researchers frequently fix this parameter across models without sufficient justification. However, embedding size is a central hyperparameter that can substantially affect model capacity and performance. This is particularly important for matrix-factorization-based approaches, in which user and item embeddings constitute the primary trainable parameters [56, 61]. Consequently, restricting the embedding dimension to a fixed or relatively small value may disadvantage certain models and lead to misleading performance comparisons [56, 62]. Several studies [63, 64] have shown that models are sometimes optimized using test data, introducing data leakage and producing overly optimistic performance estimates. In addition, stochastic models may exhibit considerable variation across different random seeds. Evidence from computer vision [65], for example, demonstrates that the choice of random seed can substantially influence reported performance, highlighting the importance of accounting for such variability when assessing empirical improvements.

Although these methodological issues have been recognized for more than a decade, it remains unclear whether they persist in recent research. Therefore, this thesis work aims to: (i) reproduce recent findings for top-n, SBRS, and IARS models; (ii) identify methodological issues, such as optimization on test data, unjustified restrictions on embedding size, and unequal hyperparameter tuning between proposed and baseline models; and (iii) evaluate the effectiveness of reproduced top-n, SBRS, and IARS against well-tuned benchmark models from different algorithmic families. Across the six studies included in this thesis, the first four investigate reproducibility and empirical progress through controlled experiments in selected recommendation settings, whereas the fifth and sixth assess whether contemporary reporting and artifact-sharing practices enable independent reproduction at scale.

1.0.1 Contributions

The first study examines the role of hyperparameter optimization in RS evaluation. We reviewed 21 top-n recommendation papers published in the 2022 proceedings of KDD, RecSys, SIGIR, The Web Conference, and WSDM and found that tuning procedures were generally insufficiently documented, particularly for baseline models. Only two papers reported both the search ranges and optimal hyperparameters for all models and datasets, while many studies fixed the embedding size or reused default values from earlier work. To illustrate the consequences, we compared fine-tuned and untuned versions of seven recent models. From this illustrative experiment, the most important observation was that, for each dataset, the worst-performing tuned model substantially outperformed the best-performing untuned model. This finding may not be unexpected, given that the performance of neural recommendation models strongly depends on the selected hyperparameter values. However, our literature review revealed limited reporting of

hyperparameter-tuning procedures, particularly for baseline models. Consequently, without access to the considered hyperparameter ranges and, more importantly, the optimal values identified for each dataset, verifying the claims made in published studies becomes extremely difficult, if not impossible.

Building on this methodological perspective, the second study evaluates the performance of GNN-based models for SBRS. We reviewed recent publications from high-quality venues and identified 34 relevant studies. From these, eight GNN-based models could be reproduced with reasonable effort using the original author-provided implementations. To ensure a fair comparison, all models were integrated into a common evaluation framework, systematically tuned, and evaluated under identical conditions on three widely used e-commerce datasets. We additionally included four simpler baselines based on nearest-neighbour and sequential rule-based techniques. The results showed that the simpler models outperformed all GNN-based approaches in terms of Mean Reciprocal Rank (MRR), which was also used as the hyperparameter-optimization criterion. However, GNN-based models achieved higher Hit Rate (HR) values in three experimental settings. We further examined the effects of embedding size, random-seed selection, and hyperparameter tuning on the test data. These experiments demonstrated that all three factors can substantially influence reported performance and may lead to unreliable conclusions.

The third study extends this investigation to IARS and evaluates the reproducibility of their reported results. We initially identified 88 publications related to intent-aware recommendation and subsequently retained 13 recent top-n studies published in highly ranked venues. To assess reproducibility, we adopted an evaluation strategy based on the original experimental configurations, including the author-provided code, datasets, data splits, and reported hyperparameters. With reasonable effort, we were able to execute five models; however, the reported results in the papers could be reproduced only for a subset of them. We then compared these models with systematically tuned traditional baselines. In every case, at least one traditional baseline matched or outperformed the examined intent-aware model, despite not explicitly modeling users' latent intents. Moreover, the original studies did not include these well-known traditional methods in their comparisons and instead focused largely on recent neural approaches. These findings indicate that restricted baseline selection and comparisons within a single algorithmic family may lead to a limited progress.

Building on the first three studies, which identified methodological weaknesses in top-n, SBRS, and IARS, the fourth study further examines whether recent GNN-based SBRS models consistently outperform established neural baselines. In this work, we compared six recent GNN approaches with three established sequential neural models: GRU4Rec, NARM, and STAMP. All models were evaluated under identical conditions using their official implementations, systematic hyperparameter tuning, and three commonly used datasets. The results showed that GRU4Rec achieved the best performance on two datasets, while the earlier SR-GNN model performed best on the third dataset. Our findings again point to the persistence of methodological weaknesses in recent models that claim improvements over existing baselines.

The fifth study then broadens the scope from individual model families to the wider RS literature by assessing whether current reporting and artifact-sharing practices are sufficient to reproduce complete experiments and verify claimed advances over the state of the art. In this study, we began our investigation by examining reproducibility practices in RS research. We analyzed 65 algorithmic papers published at ACM SIGIR (2022-2024) and assessed the availability of source code, datasets, preprocessing procedures, data splits, hyperparameter information, evaluation details, and supporting artifacts. Our findings were concerning. While the source code of proposed models was shared in 75% cases, but for baseline models, it was provided only in eight cases. Hyperparameter tuning of baseline models was very sparse and none of the examined papers documented how they were actually determined. Consequently, the full experiments and the reported claims of progress were generally difficult or impossible to reproduce. This study therefore highlights the need for more complete artifact sharing, transparent baseline optimization, and stronger reproducibility standards.

The last study extends the fifth study by examining reproducibility challenges specific to large language model (LLM)-based recommender systems. It investigates both general reproducibility issues and LLM-specific factors that hinder their reproducibility, including model versioning, prompt documentation, fine-tuning materials, tokenizer specifications, context length, and temperature settings.

In the tables below, we provide the metadata for each publication and describe the contributions of the individual authors, including the distribution of tasks involved in the completion of each study.

Paper 1	
Title	Everyone’s a Winner! On Hyperparameter Tuning of Recommendation Models
Authors	Faisal Shehzad and Dietmar Jannach
Conference	ACM RecSys 2023
DOI	https://dl.acm.org/doi/10.1145/3604915.3609488
Code, data, and documentation	https://faisalse.github.io/RecSys2023_hyperparameter_tuning/
Distribution of Tasks	The research objectives were jointly defined and addressed by both authors. I was responsible for preparing the data, setting up the experiments, and validating the experimental pipeline with support from my co-author. My co-author led the writing of the manuscript, while I contributed to reviewing the manuscript and preparing the relevant online materials.
Contribution	(i) Analyzed how hyperparameter tuning of baseline models is documented in RS papers published at top-ranked conferences, (ii) conducted an illustrative experiment comparing systematically tuned models with their untuned counterparts, demonstrating that improper or selective hyperparameter tuning can make almost any model appear to be an “arbitrary winner,” and (iii) showed that simpler recommendation models can achieve competitive or superior performance when properly tuned, emphasizing the importance of including strong and well-optimized simple baselines in experimental comparisons.
Reference	Shehzad, F., and Jannach, D. (2023). Everyone’s a Winner! On Hyperparameter Tuning of Recommendation Models. In <i>Proceedings of the 17th ACM Conference on Recommender Systems</i> .

Paper 2	
Title	Performance Comparison of Session-Based Recommendation Algorithms Based on GNNs
Authors	Faisal Shehzad and Dietmar Jannach
Conference	ECIR 2024
DOI	https://doi.org/10.1007/978-3-031-56066-8_12
Code, Data, and Documentation	https://faisalse.github.io/SessionRecGraphFusion/
Distribution of Tasks	The research objectives were jointly defined and addressed by both authors. I prepared the data, designed and conducted the experiments, developed the online supplementary materials, and wrote the initial draft of the manuscript. My co-author contributed to revising and reviewing the manuscript and improving the overall quality of the publication.
Contribution	(i) Integrated eight recent GNN-based session-recommendation models published in highly ranked venues into the session-rec framework, (ii) conducted an independent and reproducible comparison of these GNN-based session-recommendation models with four simpler non-neural models based on k-nearest-neighbour and sequential rule-based approaches, (iii) demonstrated that properly tuned simpler models can match or outperform advanced GNN-based models, and (iv) conducted additional robustness experiments to examine methodologically problematic evaluation practices and sources of instability, including tuning hyperparameters on the test data, fixing the embedding size across models, and relying on a single random seed.
Reference	Shehzad, F., and Jannach, D. (2024). Performance Comparison of Session-Based Recommendation Algorithms Based on GNNs. In <i>Advances in Information Retrieval: Proceedings of the European Conference on Information Retrieval</i> .

Paper 3	
Title	A Worrying Reproducibility Study of Intent-Aware Recommendation Models
Authors	Faisal Shehzad, Maurizio Ferrari Dacrema and Dietmar Jannach
Conference	ACM SIGIR 2025
DOI	https://doi.org/10.1145/3726302.3730307
Code, Data, and Documentation	https://faisalse.github.io/IntentAwareRS/
Distribution of Tasks	The research objectives were jointly defined and addressed by Dietmar Jannach and me. I prepared the data, designed and conducted the experiments, developed the online supplementary materials, and wrote the initial draft of the manuscript. Dietmar Jannach contributed to the experimental design and supported the writing and revision of the manuscript. Maurizio Ferrari Dacrema contributed to improving the overall quality of the manuscript and reviewed the online materials to identify previously unnoticed errors and ensure that the experiments were easy to reproduce.
Contribution	(i) Reproduced five neural intent-aware recommendation models published in highly ranked academic venues using the source code, datasets, and experimental configurations provided by the original authors, and (ii) benchmarked these models against traditional top-n recommendation approaches, including k-nearest-neighbour, graph-based, and linear models.
Reference	Shehzad, F., Ferrari Dacrema, M., and Jannach, D. (2025). A Worrying Reproducibility Study of Intent-Aware Recommendation Models. In <i>Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval</i> .

Paper 4	
Title	Revisiting the Performance of Graph Neural Networks for Session-Based Recommendation
Authors	Faisal Shehzad and Dietmar Jannach
Conference	ACM RecSys 2025
DOI	https://doi.org/10.1145/3705328.3748156
Code, Data, and Documentation	https://faisalse.github.io/RevisitGNN_SessionRec/
Distribution of Tasks	The research objectives were jointly defined and addressed by both authors. I prepared and preprocessed the data, developed the experimental pipeline for executing and evaluating all considered algorithms, prepared the online repository containing the source code and supplementary materials, and wrote the initial draft of the manuscript. My co-author contributed to the writing and critical review of the manuscript, thereby improving its clarity and overall quality. He also reviewed the online repository to enhance its reproducibility, usability, and ease of execution.
Contribution	(i) Evaluated six recent GNN-based SBRS published in highly ranked academic venues using the official implementations provided by their original authors, (ii) benchmarked the GNN-based models against three established sequential neural models: GRU4Rec, STAMP, and NARM, and (iii) systematically tuned the hyperparameters of all sequential neural and GNN-based models for each considered dataset and evaluated them under identical experimental conditions within a unified evaluation framework.
Reference	Shehzad, F., and Jannach, D. (2025). Revisiting the Performance of Graph Neural Networks for Session-Based Recommendation. In <i>Proceedings of the ACM Conference on Recommender Systems</i> .

Paper 5	
Title	“We Share Our Code Online”: Why This Is Not Enough to Ensure Reproducibility and Progress in Recommender Systems Research
Authors	Faisal Shehzad, Timo Breuer, Maria Maistro and Dietmar Jannach
Conference	ACM RecSys 2025
DOI	https://doi.org/10.1145/3705328.3748157
Data and Documentation	https://faisalse.github.io/SIGIR-reproducibility-audit/
Distribution of Tasks	The research objectives were jointly defined and addressed by all authors. I collected and analyzed the data and presented the findings to my co-authors for discussion and interpretation. Dietmar Jannach prepared the main draft of the manuscript. The other co-authors and I contributed to revising and critically reviewing the manuscript, thereby improving its clarity and overall quality. Timo Breuer and I prepared the online supplementary materials. Following acceptance, I implemented the necessary revisions to the manuscript and supplementary materials with the approval of the other co-authors.
Contribution	(i) Systematically selected and analyzed 65 RS papers published at ACM SIGIR between 2022 and 2024, and (ii) developed and applied a structured set of reproducibility variables to assess the availability and documentation of research artifacts, including source code, datasets, data splits, hyperparameter configurations, evaluation procedures, and supporting documentation.
Reference	Shehzad, F., Breuer, T., Maistro, M., and Jannach, D. (2025). “We Share Our Code Online”: Why This Is Not Enough to Ensure Reproducibility and Progress in Recommender Systems Research. In <i>Proceedings of the Nineteenth ACM Conference on Recommender Systems</i> .

Paper 6	
Title	Reproducibility of LLM-based Recommender Systems Research: Worrying Observations from an Initial Analysis
Authors	Faisal Shehzad and Dietmar Jannach
Conference	ACM UMAP 2026
DOI	https://doi.org/10.1145/3774935.3812708
Data and Documentation	https://faisalse.github.io/LLM-reproducibility-audit/
Distribution of Tasks	The research objectives were jointly formulated and addressed by both authors. I collected and analyzed the data and presented the findings to my co-author for discussion and interpretation. I prepared the initial draft of the manuscript, while my co-author contributed to its review and revision to improve the overall quality. After acceptance, I implemented the required revisions to the manuscript and supplementary materials with the approval of my co-author.
Contribution	(i) Systematically selected and analyzed 64 recent LLM-based recommender-system papers published at top-ranked conferences, and (ii) designed and checked LLM-specific reproducibility variables to assess aspects such as model and version specification, prompt documentation, fine-tuning materials, tokenizer information, context length, temperature settings, model checkpoints, and other artifacts required to reproduce LLM-based recommendation experiments.
Reference	Shehzad, Faisal, and Dietmar Jannach. "Reproducibility of LLM-based recommender systems research: Worrying observations from an initial analysis." Proceedings of the 34th ACM Conference on User Modeling, Adaptation and Personalization. 2026.

1.1 Scope and limitations

This thesis examines the reliability, reproducibility, and validity of experimental evaluation in RS research. In particular, it investigates comparisons between simpler and more complex models, the influence of hyperparameter tuning and other methodological choices, and the availability of reproducibility artifacts required to ease the reproducibility of reported results in a paper. The empirical studies are limited to offline evaluations of selected top-n, SBRS and IARS using publicly available benchmark datasets. The reproducibility analysis focuses on studies published in selected leading conferences and journals. Although these selections do not represent the entire field of RS research, they provide a sufficiently strong basis for identifying recurring patterns and drawing well-supported conclusions within the defined scope of this thesis.

1.2 Key takeaways

In this section, we present the main takeaways of our research work:

1. Unequal hyperparameter tuning can produce arbitrary model rankings. Extensively tuning a proposed model while using default or previously reported settings for baselines may lead to unfair comparisons and unreliable claims on selected benchmark models.
2. Many of the examined studies fix the embedding size, even though it is a hyperparameter that should be tuned appropriately to ensure fair comparisons among the evaluated models.
3. The analysis of the selected studies indicates that newly proposed models are often compared only with recent algorithms or with models from the same methodological family, which may lead to unfair comparisons.
4. Long-established, simpler models have demonstrated competitive performance in numerous empirical studies. However, most of the examined studies either did not include these models as baselines or evaluated only weak or insufficiently tuned versions of them.
5. The choice of random seeds may influence experimental results. Therefore, where computationally feasible, the selected models should be executed multiple times using different random seeds, and the average results should be reported.
6. A few examined studies exhibited potential data-leakage issues, such as tuning models on the test set or allowing training samples to appear in the test data. Such practices can lead to biased and unreliable performance estimates. Therefore, data-splitting procedures should be carefully verified, and model selection and hyperparameter tuning should be performed exclusively on the validation set.
7. To support full reproducibility, the authors should release a complete reproducibility package containing the source code for the proposed and baseline models, hyperparameter tuning scripts, data-preprocessing code, dependency files specifying the exact

software environment, and detailed instructions for reproducing the experiments on each dataset used in a paper.

1.3 Future directions

In future work, we will explore two research directions that are currently reshaping RS research and have been identified in numerous studies as promising approaches for improving reproducibility.

1.3.1 Reproducibility of Large language model-based recommender systems

Large language models (LLMs) have emerged as powerful tools in natural language processing and have attracted substantial attention in RS research. These models are trained on large-scale and heterogeneous data and can learn rich semantic representations that may support the modeling of users, items, preferences, and contextual information. In recent years, LLMs have increasingly been incorporated into RS for tasks such as user and item representation learning, preference elicitation, recommendation generation, and explanation. However, their use introduces additional reproducibility challenges. In future work, we will therefore examine the reproducibility of LLM-based RS, focusing on the effects of model and API versioning, prompt specification, stochastic generation settings, fine-tuning procedures, training and evaluation data, shared embeddings, and dependencies on proprietary services. Particular attention will be paid to whether the reported experimental results can be independently reproduced when the underlying models or APIs are updated, or available with restricted access or no longer publicly accessible.

1.3.2 Reliability and reproducibility of third-party recommender system frameworks

RS research increasingly relies on reusable evaluation frameworks to facilitate model implementation, experimentation, and evaluation. Frameworks such as Elliot, DaisyRec, and RecBole can promote reproducibility and transparency by providing implementations of benchmark algorithms, automated procedures for hyperparameter optimization, standardized data-processing pipelines, support for publicly available datasets, and implementations of commonly used evaluation metrics. In recent years, such frameworks have therefore become increasingly important for improving the consistency and reproducibility of experimental research. In [62], we analyzed 65 papers published at a top-ranked venue and found that 14 papers, corresponding to approximately 21% of the sample, reported using an existing evaluation framework. Despite this development, reproducibility in RS research remains difficult to achieve. Previous work has shown that third-party implementations may differ from official author-provided implementations and may therefore produce inconsistent or unreliable results. In addition, [66] identifies issues related to the implementation of early stopping in the RecBole framework. Building on these observations, our future study will systematically examine the reliability and consistency of reusable RS frameworks. Specifically, we will investigate their implementations of integrated recommendation models,

data-preprocessing and splitting procedures, automated hyperparameter-optimization mechanisms, early-stopping criteria, and evaluation-metric computations. We will also compare model performance across frameworks under equivalent experimental configurations and benchmark the framework implementations against the official implementations released by the original model authors.

1.4 Conclusion

Overall, this thesis shows that increasing model complexity does not necessarily translate into reliable progress in RS research. Across top-n, SBRS and IARS, our studies demonstrate that well-tuned traditional and established models can remain highly competitive with, and in several cases outperform, more recent complex approaches. The findings further highlight how baseline selection, hyperparameter optimization, embedding size, test-set tuning, random seeds, and incomplete artifact sharing can substantially influence reported results. Collectively, these studies emphasize the need for stronger baselines, systematic and transparent optimization, consistent evaluation procedures, and complete reproducibility artifacts to support credible and verifiable progress in RS research.